

KATHIRVEL A

EMBEDDED DEVELOPER

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CAREER OBJECTIVES

I am an Embedded systems enthusiast with strong hands-on experience in STM32, ESP32 and bare-metal Embedded C. Seeking roles in product development and Embedded systems engineering .

TECHNICAL SKILLS

- Programming Languages: C, Embedded C, C++, Python
- Embedded Concepts: Bare-Metal Programming, Interrupt Handling
- Real-Time system: FreeRTOS
- Communication Protocols: UART, SPI, I2C
- MCU Peripherals: ADC, DAC, Timers, PWM
- Microcontrollers: STM32 (ARM Cortex-M4), ESP32, Arduino
- Robotics and Middleware: Robot Operating System (Learning)
- Tools: STM32CubeIDE, Arduino IDE, VS Code, Fusion 360
- Operating System: Linux (Ubuntu), Windows
- Computer Vision: Hand Detection, Pose Detection, Face Detection (MediaPipe)

INTERSHIP

- **Elamani Tech Pvt Ltd**
 - Analyzed and studied the embedded hardware architecture of a commercial JingWu service robot, including controller boards, sensors, power distribution, and system-level integration.
 - Developed bare-metal STM32 firmware for the CarLupar intelligent pan-tilt lighting system, implementing ADC-based joystick control, I2C communication, PCA9685 PWM generation, and servo motor control.

PROJECTS

- **DataLogger Using STM32:**

Developed an STM32-based data logger to acquire GPS (USART) and analog sensor (ADC) data with interrupt-driven processing. Stored data in W25Q64 SPI flash and enabled real-time display via UART communication.

- **AI Vision-based Wireless Robot:**

Built a computer vision-based robot that detects real-time hand gestures using MediaPipe. Integrated Arduino with Bluetooth communication to enable wireless gesture-controlled movement.

- **ExpiraTrack – Smart Expiry Monitoring Website**

Developed an AI-powered expiry tracking system using Generative AI tools for intelligent date monitoring. Designed to automate alerts and improve inventory efficiency with smart tracking logic.

LEADERSHIP & ACHIEVEMENTS

- Served as class representative from the second semester to the present, acting as a liaison between students and faculty and coordinating academic communication and class activities.
- Won 1st Place – Overall Best Mini Project among all departments at college-level competition
- Achieved 1st Place in Infinity Quest and 3rd Place in Neuronumerics at a symposium conducted by the Department of Mathematics, St. Joseph's College.